

STRATEX™ FORENSIC ENVELOPE AUDIT · V1.0

Strategic Thermal Reconnaissance Report

142 Bluegrass Ridge · Lexington, KY · Scan 2026-06-24 · Ground-truth accuracy ±2.5 cm

TOTAL SQUARES 28.00 SQ (100 sf)	SHINGLE LAYERS 1 / 2 DETECTED / CODE MAX	FACETS 10 DISTINCT PLANES	ANOMALIES FLAGGED 3 FORENSIC HOTSPOTS
EXECUTIVE SUMMARY · GRAND TOTAL RANGE \$10,600.00 → \$12,200.00 MATERIALS \$6,540.00 · LABOR \$3,360.00 · UNFORESEEN \$700.00-\$2,300.00		STRATEX RECOMMENDATION TEAR OFF No significant standing water detected on primary roof planes; minor localized pooling possible in gutters due to debris.	

Building Envelope Score

ENERGY DEFICIENCY

30%

DEFICIENT

VENTILATION

60%

POOR

OVERALL ENVELOPE

65/100

COMPOSITE

AI-PRIORITIZED MAINTENANCE QUEUE

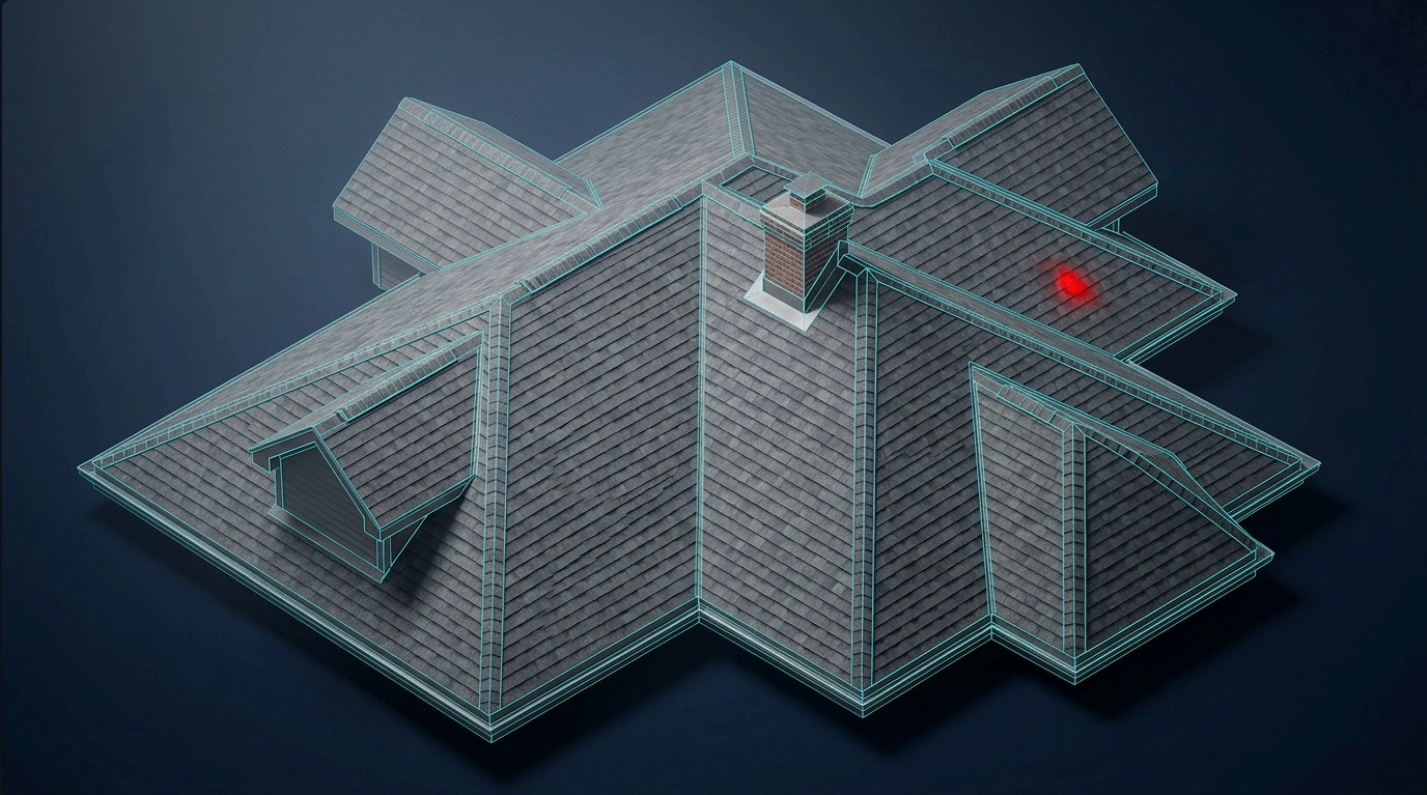
1	URGENT Full Roof System Replacement	\$350/yr
2	HIGH Improve Attic Ventilation (Ridge Vent System)	\$180/yr
3	MED Address Deteriorated Flashing at Chimney	\$0/yr

THERMAL FORENSIC FINDINGS

Attic Heat Loss	+5.2°C above ambient	HIGH
HVAC Duct Leak (Garage)	+3.8°C above ambient	MED

Sub-surface Anomaly Atlas

RADIOMETRIC OVERLAY • ROOF



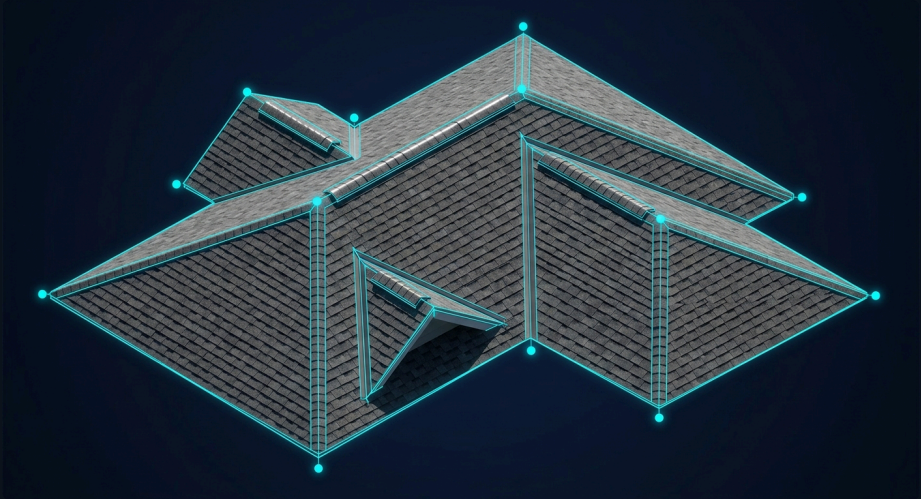
CYAN: scan-vector overlay • ORANGE: thermal hotspot • Confidence threshold ≥ 85%

ANOMALY LEDGER

SEV	ID	TYPE	LOCATION	CONF	SF	DIAGNOSIS	EST
MED	A1	Granule Loss	North Slope	95.5%	5.0	Age-related wear, UV exposure	\$300.00
LOW	A2	Minor Shingle Buckling	West Slope	92.0%	2.0	Thermal stress, possible inadequate ventilation	\$150.00
MED	A3	Flashing Deterioration	Chimney Perimeter	94.8%	1.0	Sealant failure, material fatigue	\$400.00

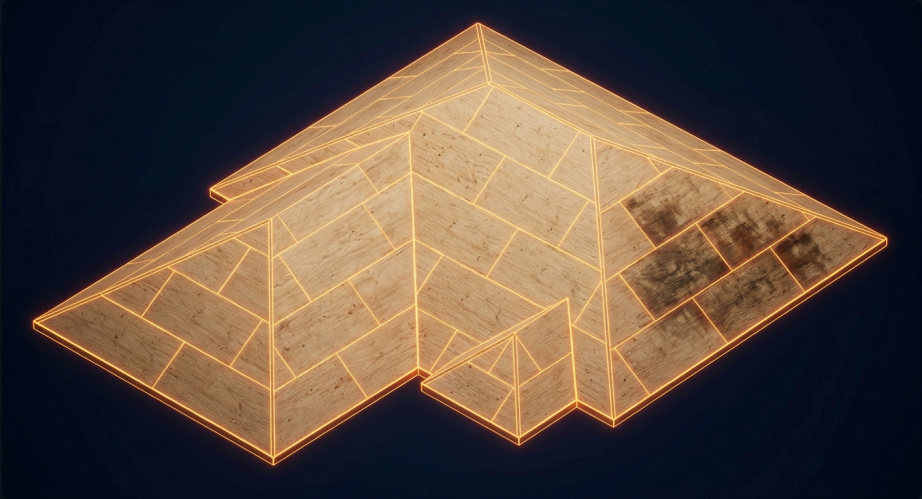
Layered CAD/BIM Twin

LAYER 01 · FINISHED SLATE



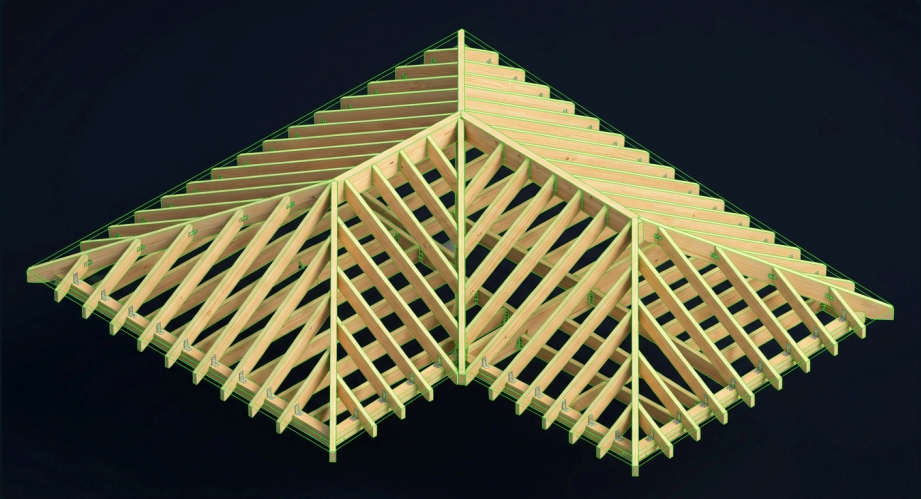
Squares 28.00 · Pitch 8/12

LAYER 02 · DECKING + UNDERLAYMENT



7/16" CDX · I&W shield at valleys/eaves · 1 existing shingle layer(s)

LAYER 03 · STRUCTURAL FRAMING



2x8 rafters @ 16" o.c. · LVL ridge · Hurricane ties **VERIFIED**

VALLEYS

50.0

LINEAR FT

GABLES

80.0

LINEAR FT

RIDGES

35.0

LINEAR FT

EAVES

180.0

LINEAR FT

Fenestration Schedule

ID	LOCATION	SHAPE	W	H	QTY	U-FACTOR	LEAK SEV
W1	Front Elevation	Double-hung	36"	60"	4	0.35	LOW
W2	Side Elevations	Casement	24"	48"	6	0.38	NONE
W3	Rear Elevation	Picture	72"	48"	1	0.33	NONE

Exterior DoorSchedule

ID	LOCATION	SHAPE	W	H	QTY	WEATHERSTRIP
D1	Front Entry	Arch-top Panel	36"	80"	1	WORN
D2	Rear Patio	Sliding Glass	72"	80"	1	OK

Bill of Materials

CATEGORY	MATERIAL	SKU	QTY	UNIT	UNIT COST	LINE TOTAL
Roofing	Architectural Shingles	GAF Timberline HDZ	90	bundles	\$45.00	\$4,050.00
Underlayment	Synthetic Felt	GAF Deck-Armor	3	rolls	\$100.00	\$300.00
Waterproofing	Ice & Water Shield	GAF StormGuard	2	rolls	\$150.00	\$300.00
Roofing	Starter Shingles	GAF QuickStart	3	rolls	\$70.00	\$210.00
Roofing	Hip & Ridge Cap	GAF TimberTex	4	bundles	\$80.00	\$320.00
Ventilation	Ridge Vent	GAF Cobra Attic Vent	4	sections (10ft)	\$60.00	\$240.00
Perimeter	Aluminum Drip Edge	Standard	26	pieces (10ft)	\$10.00	\$260.00
Penetrations	Pipe Boots	Oatey Master Flash	4	units	\$25.00	\$100.00
Consumables	Sealant / Caulk	Roofing Grade	1	case	\$80.00	\$80.00
Fasteners	Roofing Nails	1.25" Galvanized	6	boxes	\$30.00	\$180.00
Waste Management	Dumpster Rental	20-yard	1	unit	\$500.00	\$500.00
MATERIALS TOTAL						\$6,540.00

Crew

Economics

RATE BENCHMARK		CREW PLAN		
NATIONAL AVG / SQ	REGIONAL AVG / SQ	CREW SIZE	DAYS	LABOR TOTAL
\$175.00	\$160.00	5	2.5	\$3,360.00
REGION: Midwest • APPLIED: \$120.00/SQ				

08 · CONTINGENCY ENVELOPE

Unforeseen Repair Reserve

ITEM	PROB	LOW	HIGH	NOTE
Plywood Decking Replacement	30%	\$500.00	\$1,500.00	For any rotten or delaminated decking discovered after tear-off (est. 5-15 sheets).
Fascia/Soffit Repair	20%	\$200.00	\$800.00	For localized damage to fascia or soffit boards at eaves or gables.
RESERVE RANGE		\$700.00	\$2,300.00	